
Utility-Certified Songline Memory: Consolidation as a Two-Axis Decision, Learned and Evolved

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Abstract

Problem. The preceding parts of this series closed the *exploitation and coordination* of already-formed distributed memory (provenance contracts, transfer units, merge/trust/staleness rules). They left open its *ontogenesis*: how agents learn, in the course of joint activity, what deserves to be remembered, what becomes a landmark, and what fragment of experience becomes a song. **Model.** We define memory formation as a bounded choice among five operations — merge, exception, new schema, repeat, drop — driven by two *independent* axes: counterfactual marginal utility $U(m \mid M) = \text{cost}(M) - \text{cost}(M \cup \{m\})$, measured by deterministic replay with the memory switched on and off, and simplicity of structural analogy (signature overlap plus *decision distortion* — where two songs ultimately send the walker). Stored schemas carry *utility certificates* (conditions, observed gain, per-role utility profile, evidence, failures); receivers recompute their own utility rather than trust the sender’s number. **Findings.** A deterministic prototype passes all four registered predictions: duplicates cost nothing by construction, the exception operation beats both last-write-wins merging (12.3 vs. 62.6 steps on the corrupted world) and recency shadowing, and similarity is measurably not a consolidation criterion. A contextual bandit *learns* the decision matrix from coordination feedback alone and surpasses the hand-designed thresholds (78.1 vs. 83.5 vs. random 105.9); its one principled deviation — append instead of overwrite — is exactly what pays. An evolution loop over the song grammar selects a *finite* couplet budget with no cost regression and converges to the neighbourhood of the hand-designed thresholds. A seven-policy comparison over heterogeneous embodiments at 10/100/1000 episodes ends in four registered failures whose mechanisms are the finding: structural memory beats embedding-similarity, recency-importance-relevance and full-snapshot baselines by $1.7\text{--}4.5\times$ at every scale; consolidation buys *bits*, not success (154k vs. raw’s 339k vs. snapshots’ 5.2M at 1000 episodes); and the long-horizon failure mode is a *staleness* problem, with the composition ladder measured: consolidation -14% , $+freshness -3\%$, $+trust\text{-}flip -0.9\%$, closing the robust role against raw context. **Later waves.** The landmark-promotion loop closes causally over a candidate superset (deliberately harmful junk caught at $\Delta = -0.67$; a zero-attribution border feature proved causally load-bearing — attribution and causal necessity are different quantities); a cross-family signature codebook monetises recurring motifs (-16% bits at 1000 episodes, growing with scale); a graph-matching analogy engine cuts memory 24% at equal cost; terminal-only meta-learning beats both the densely-shaped bandit and the hand matrix; a live-LLM end-to-end cycle shows reconstruction beating replay — raw transcripts degrade from 0.42 to 0.08 (llama3.1) and from 1.0 to 0.375 (Qwen2.5-3B) as conflict history grows, while the reconstructed active

context holds 1.0 at 6–12% of the payload; and the social experiment returns the programme’s strongest honest negative: at long horizon in a globally drifting world, independent agents beat every exchange arm — broadcast time is not world time, the series’ gate-not-exchange law re-derived at the formation level. The social frontier then closes along a measured causal ladder: unregulated exchange is harmful (S1); a world-time staleness contract is necessary but insufficient (S2); and receiver-validated admission — accept foreign memory only where it has shown utility on one’s own visits — makes the collective beat independence on every seed and both roles (S3, −20%). Finally all mechanisms compose into one runtime that passes a reviewer’s acceptance test: a runtime utility estimator matching the exact counterfactual ($\rho=0.989$) so no oracle is needed; equal-budget dominance where raw history does not catch up at $4\times$ the memory; and safety gates driving fail-open below 1% on both a grid and a continuous substrate. **Thesis.** What to remember is decided not by descriptive salience but by counterfactual usefulness for future individual and collective decisions, corrected by the cost of structural transfer between agents’ local worlds — and both the decision rule and the grammar it operates on is learnable, its parameters evolvable within a fixed grammar family. **Scope.** Grid substrates; deterministic replay; a measurement programme with registered predictions, not a general theory.

1 Introduction: from coordination to ontogenesis

The series so far treats memory that already exists. Provenance annotation prices foreign evidence at the moment of commitment; the song transfers routes and identity without shared frames; the merge/trust/staleness triple makes a collective of local memories coordinate. What none of these mechanisms does is *form* the memory: decide which observation becomes a record, which record becomes a schema, which feature earns landmark status, and what is worth broadcasting at all. The single sharpest single-agent diagnosis of the series — candidate generation starves at 91% of decision points while ranking is near-optimal — is precisely a formation problem, and the open problem list of the ancestor manuscript names selective consolidation as its direct attack.

This part states the formation problem as a bounded decision, gives it a measured deterministic solution, and then subjects both the decision rule and the grammar beneath it to learning and selection. Throughout, one methodological commitment from the series carries over unchanged: predictions are registered before runs, and failures against registration are reported with their mechanisms — several of the most informative results below are registered failures.

The key idea. Usefulness and simplicity-of-analogy are two *independent* axes, and no single scalar (novelty, similarity, importance, reward) can substitute for their product structure:

Utility	Analogy	Memory operation
high	simple	MERGE into the existing schema
high	simple but decision-conflicting	EXCEPTION: a counterexample, never an overwrite
high	complex	NEW_SCHEMA
low	simple	REPEAT: update statistics, no new record
low	complex	DROP

The fifth cell is the load-bearing one: a useful event that contradicts an existing schema can be neither merged (it would corrupt) nor dropped (it is evidence); it must be stored as an exception until the schema splits. This is what protects the memory from false generalisation.

2 The model

Counterfactual marginal utility. For memory item m , agent i , intent ι and current memory M :

$$U_{i,\iota}(m \mid M) = \text{cost}_i(M, \iota) - \text{cost}_i(M \cup \{m\}, \iota),$$

measured by *deterministic replay* with the item switched on and off — the source-masked replay device of the warp analysis, generalised from “foreign evidence on/off” to “this memory on/off.”

Utility is contextual (per agent and intent) and marginal (relative to what is already stored): a duplicate of a stored schema has $U = 0$ by construction, so repeats are detected without any similarity heuristic. The team-level version subtracts collision, communication and false-transfer costs.

Analogy cost. A new episode e maps onto a stored schema s through ϕ with cost

$$C_{\text{analog}}(e, s) = L(\phi) + \lambda_d D_{\text{decision}}(e, s) + \lambda_a \text{Defect}(\phi), \quad A(e, s) = e^{-C_{\text{analog}}},$$

where L counts structure outside the correspondence (here: couplets outside the signature LCS), D_{decision} measures how differently the two songs *decide* — operationalised as the distance between their end-to-end beat displacements, i.e. where they ultimately send the walker — and Defect penalises broken relational structure. Descriptive similarity and decision agreement are different quantities, and D_{decision} captures the second; this is the rate-distortion intuition of Anonymous (2026a) imported into a multi-agent, structural setting.

The memory unit is a conditional programme, not a fact: $m = (G_m, C_m, E_m, U_m, \Phi_m, P_m, F_m)$ — relational graph/song, applicability conditions, expected effects, a utility *profile by role*, known analogical maps, provenance with references to immutable source episodes, and known failures. On exchange, a schema travels with a **utility certificate** (conditions, ΔV , uncertainty, evidence ids, analogy map, failures); the receiver transports the song through its own mapping and recomputes its own utility. For heterogeneous embodiments this is essential: a route through a hazard field can have $U_{\text{robust}} > 0$ and $U_{\text{fragile}} < 0$ simultaneously, so a memory item carries a utility function over roles, not one weight.

Four storage layers. (1) an *immutable episodic store* — source episodes are never rewritten (continuous rewriting first helps and then hurts (Anonymous, 2026d); primary evidence anchors reconsolidation); (2) the *analogical schema graph* — songs, landmarks, exceptions, governed by the two-axis matrix; (3) a *procedural layer* for schemas promoted to policy fragments; (4) a *coordination layer* for social rules (reservations, roles, trust, transfer conditions). An agent’s active context is a reconstruction — relevant schemas, minimal supporting episodes, current exceptions and the coordination contract — not a replay of history.

2.1 The runtime as one algorithm

Every experiment below is a configuration of one per-agent loop; we state it once so the method is reconstructable from the paper alone. A record is $m = (G, C, E, U, P, T, R, F, A)$ — song graph, applicability, expected effect, per-role utility, provenance, observation/world time, frame correspondences (re-derived at consumption, never stored), exceptions, and admission/authority state.

Algorithm 1 (Songline Memory Runtime).

In: observation o , intent ι , local memory M , quarantine Q .

1. extract candidate song from o ; 2. estimate marginal utility $\hat{U}$ (§10, UE1); 3. find structurally closest schema (analogy §8); 4. compute analogy cost + decision distortion;
5. choose DROP / REPEAT / MERGE / EXCEPTION / NEW_SCHEMA; 6. append to immutable episodic store; 7. update schema graph; 8. build provenance + utility certificate;
9. broadcast on cadence, within wire budget; 10. place received records in Q ; 11. on visit, validate world-time, role, local utility; 12. promote Q -record to retrieval/planning/action authority; 13. retrieve an admissible song (support $\times$ staleness order); 14. recover landmark correspondence (consensus $\geq k + \text{loop closure}$); 15. reserve target/route; 16. execute safe prefix (≥ 5 couplets verified); 17. detect rupture; 18. commit / replan / rollback.

Quantities in one place. Marginal utility $U_{i,\iota}(m|M) = \text{cost}_i(M, \iota) - \text{cost}_i(M \cup \{m\}, \iota)$; analogy cost $C = L(\phi) + \lambda_d D_{\text{decision}} + \lambda_a \text{Defect}(\phi)$; retrieval score $\text{support} \cdot e^{-\alpha_{\text{age}}}$ (trust-flip $\times 0.25$ on an excepted parent); world-clock admissibility $\text{ver}(m) \geq \text{known_ver}(\text{fam})$; rupture index = $\arg \min_e \text{trust} \cdot e^{-\alpha_{\text{age}}(e)} < \tau$; objective $\alpha \text{succ} + \beta(-t) - \gamma \text{bits} - \delta \text{collisions}$.

Hyperparameters (frozen; test seeds never used to tune). utility threshold $U_{\text{thr}}=5$ (validation, U1); analogy share ≥ 0.4 and distortion ≥ 3 (validation, U1); staleness $\alpha=0.05$, $\tau=0.30$ (fixed, Part II); song budget $v_{\text{max}}=10$ (*evolved*, U3); anchor consensus $k=3$ and safe-prefix depth 5 (safety design, N1/C1). Selection method and per-parameter sensitivity are tabulated in Appendix A.

3 U1: the matrix, measured deterministically

Eight valid grid-world families, five candidate memories per family — one per matrix cell, in sequence: a useful novel song (NEW), its exact duplicate (REPEAT), a song from a structurally different world (NOVEL), a song to a goal irrelevant to the water intent (IRRELEVANT), and a song from a variant world whose water has been secretly moved (CONFLICT). Two single-axis baselines run on the same stream: similarity-only (store if novel, last-write-wins merge if similar) and utility-only (store if useful, consume newest-first). Four predictions were registered; the first registration failed on two construction defects (an “irrelevant” goal that accidentally lay en route to the water; consumers spawning inside walls), both documented, and the corrected registration passed 4/4:

- **Repeats are free by construction:** the duplicate’s marginal utility is exactly 0 in every world; final memory is 3 schemas vs. 5 for append-everything.
- **Similarity is not a consolidation criterion:** the similarity-only baseline incorporates the irrelevant song and silently corrupts the original world by overwriting its schema with the conflict variant (62.6 steps and phantom-first, vs. 12.3 **steps first-try** for the two-axis memory).
- **Exceptions beat both overwrite and recency:** with the exception stored, the variant world costs 20.5 vs. 70.4 without it, while the original stays first-try correct; utility-only’s recency order phantoms first on the original (24.0 steps).
- **Certificates are recomputable:** a receiver at a different position recovers positive utility from the certificate’s evidence in every world.

4 U2: the matrix is learnable — and improves itself

A LinUCB contextual bandit replaces the hand thresholds. Features: (U , share, D , novelty, memory size); actions: the five operations; reward: the *measured* cost change of the memory on a two-world probe (the candidate’s world and the nearest schema’s origin world — the second term is what teaches the controller not to overwrite), minus a small storage tax.

The first registration failed on two evaluation defects that are themselves findings. The final battery was circular (each policy was scored on the worlds it chose to store, making random indistinguishable from designed), and two pairs of actions were *reward-equivalent*: NEW and EXCEPTION had the identical structural consequence, as did DROP and REPEAT — and the bandit recovered exactly that quotient, confusing actions strictly *within* equivalence classes while learning the drop cell at 40/43. Bookkeeping without consequences is unlearnable. Version 2 made the battery independent and gave support a consequence (consumption order). Result:

	learned	hand matrix	random
held-out battery cost	78.1	83.5	105.9

The learned policy beats its own designer by 6.5%. Its majority actions agree with the hand matrix in 3/5 cells (a registered miss); the disagreements are two context-dependent equivalences (merging a duplicate $\equiv$ repeat; repeating junk $\equiv$ drop) and one *principled, profitable deviation*: where the hand matrix says MERGE, the controller appends a new version and lets support ordering arbitrate — it independently discovered the immutable-store principle (§2), the same lesson the consolidation literature reaches from failure analyses (Anonymous, 2026c,d).

5 U3: the grammar’s parameters are evolvable — and the constants are an attractor

An evolution strategy (30 generations $\times$ 16 genomes) selects the grammar θ = (couplet gap, share threshold, distortion threshold, utility threshold, couplet budget $v_{\max}$), with fitness measured on maps the genome never sang on (cost + λ ·bits). Both registered predictions passed: the evolved genome matches the hand grammar on hold-out within registration tolerance, and it selects a *finite* budget ($v_{\max} = 10$) with no cost regression — bounded rationality obtained by selection, not imposition. Beyond registration: selection converged to the neighbourhood of the

hand-designed thresholds (share $0.74 \rightarrow 0.46$ vs. hand 0.4; distortion 3.9 vs. hand 3) while choosing denser couplets and a lower utility bar — “store more, but shorter.” The designed constants of the matrix sit near a selection attractor rather than being arbitrary.

6 U7: seven policies, two bodies, three horizons

The decisive comparison runs one mixed episode stream (new families, exact repeats, appearance variants with resampled hazard texture over persistent walls-and-water structure, and water-moved conflicts) through seven team-memory policies, consumed by two embodiments — *fragile* (hazards cost 12 steps) and *robust* (hazards cost 1, steps cost 2) — and evaluates on the families’ latest true worlds plus fresh appearance variants never streamed. Policies: raw long context (append all, newest-first), embedding-style signature similarity, recency-importance-relevance (Park et al., 2023), full-snapshot exchange, utility-only, analogy-only, and the full two-axis memory with role-profiled certificates.

Table 1: Mean fragile-role battery cost by stream length (12 seeds) and stored bits at 1000 episodes. All four registered predictions failed; the mechanisms are the finding (see text).

Policy	10 eps	100 eps	1000 eps	bits @ 1000
raw long context	128.9	53.3	139.8	339,453
utility-only	130.5	53.6	144.6	162,107
two-axis (UCSM)	132.8	60.8	145.2	153,774
signature similarity	161.1	214.7	232.1	186,843
recency-importance-relevance	190.7	225.4	250.1	339,453
full snapshots	209.4	246.6	264.5	5,198,886

What failed, and what it means. The registered dominance, sublinear-bits, role-preservation and false-merge predictions all failed. Three mechanisms carry the content. *First*, the fail-closed identity layer makes bloat cheap in steps: stale or foreign songs simply produce no anchors and hence no targets, so the raw archive is filtered for free at consumption time and pays only in bits and matching compute. *Second*, sublinear memory growth is impossible without cross-family abstraction: in an open stream where 30% of episodes found new structures, consolidation caps within-family growth ($2.2\times$ fewer bits than raw, $34\times$ fewer than snapshots) but cannot compress novelty it has no schema for. *Third*, the long-horizon failure mode — phantom-first rates near 0.95 for every song policy at 1000 episodes — is a *staleness* cascade from accumulated conflicts, which no storage policy fixes by ordering alone; it is the native territory of the series’ $\text{trust} \times \text{staleness}$ law, which none of the seven policies implemented.

The robust positives. At every scale, structural memory (identity + beats, whatever the consolidation rule) beats the mainstream heuristics — embedding similarity, recency-importance-relevance scoring (Park et al., 2023; Anonymous, 2025b), and snapshots — by $1.7\text{--}4.5\times$; and the two-axis memory is the cheapest in bits among song policies everywhere.

U7b: testing the staleness diagnosis. An eighth policy — the same two-axis consolidation consumed newest-first — was registered and run. At 100 episodes it *passes*: it matches or beats raw on both roles (52.0/67.9 vs. 53.6/68.9) at 54% of raw’s bits, closing the entire consolidation-overhead gap by ordering alone. At 1000 episodes it fails by $\sim 3\%$: under strong appearance drift, raw’s redundancy retains residual value — older variants of a song still anchor where the newest one fails to. The programme formula that survives: **consolidation buys bits, freshness buys success, and they compose at moderate horizons; the next lever is a real age gate and cross-family schemas, not better ordering.**

7 Closing three gaps: staleness, emergence, abstraction

A second registered wave attacked the three most concrete gaps the first wave localised.

214 **Staleness in consumption (U7c).** The series’ trust $\times$ staleness contract was wired into consumption:
 215 schemas ordered by support $\cdot e^{-\alpha \text{ age}}$, plus the discrete *trust-flip* — a schema that has accumulated an
 216 exception is demoted, so the counterexample is tried before its discredited parent. At 1000 episodes
 217 this closes the robust role entirely (214.2 vs. raw’s 214.3) and cuts the fragile-role gap threefold
 218 (-0.9% vs. -3% for freshness ordering alone); the strict both-roles registration still fails, honestly —
 219 residual redundancy value under appearance drift survives even the gate. The composition ladder is
 220 now measured: consolidation alone -14% , +freshness -3% , +trust-flip -0.9% .

221 **Causally promoted landmarks (E1).** The causal loop *propose* $\rightarrow$ *ablate* $\rightarrow$ *measure* $\rightarrow$ *promote*
 222 was closed: starting from a feature superset including a derived candidate and deliberately harmful
 223 frame-variant junk (cell parity), the loop caught the junk causally ($\Delta = -0.67$; its harm manifested
 224 as refusals — zero fail-open across all ablation rounds) and reached a minimal vocabulary at perfect
 225 score ($0.33 \rightarrow 1.0$). The registered expectation that content classes would survive failed in the most
 226 instructive way: the surviving pair was {wall, void} — the border feature that carries *zero* consensus
 227 mass in the attribution measurement turns out to be causally load-bearing as an *anti-aliasing guard*
 228 once the vocabulary is impoverished. Causal ablation and attribution mass are different quantities;
 229 the value of a feature is non-additive and contingent on the rest of the vocabulary.

230 **First-order cross-family abstraction (X1).** A shared signature codebook — constellations recur-
 231 ring across families stored once and referenced by index, lossless by construction — was added to the
 232 storage codec. Registered and passed: the codebook’s bits growth ($7.91\times$ per decade) is smaller than
 233 the plain codec’s ($8.42\times$), and the absolute saving grows with scale (-10% at 100 episodes, -16%
 234 at 1000): cross-family motifs exist and monetise. Sublinearity is approached, not achieved — full
 235 schema-level abstraction (slots and roles, not just shared constellations) remains the open half.

236 **The LLM end-to-end cycle (L1).** The full promised loop — observation $\rightarrow$ candidate memory
 237 $\rightarrow$ controller $\rightarrow$ schema graph $\rightarrow$ reconstructed active context $\rightarrow$ LLM action — was closed on live
 238 models (the W4 division of labour: the LLM commits the semantic target, a deterministic waypoint
 239 primitive walks). Formation is deterministic UCSM over 4- or 12-episode histories in which the
 240 water is moved once or twice, so the stale targets hold up to a 10:2 majority of the raw transcript;
 241 a fresh LLM session then commits under three context arms (no memory / full transcript / schema
 242 graph with certificates and exceptions). Registered and measured: memory is needed (the no-memory
 243 arm is 0.0 everywhere); reconstruction beats replay — llama3.1-8B degrades on the raw transcript
 244 from 0.42 to 0.08 as history grows (conflict chains $0.17 \rightarrow 0.00$) while the active context holds 1.0
 245 throughout at 6–12% of the transcript payload; Qwen2.5-3B reads the short transcript perfectly (1.0)
 246 yet drops to 0.375 on 12-episode chains, with the active context again at 1.0. A 4B thinking model
 247 fails the supersede chain in every arm — a capability floor below the task, reported for honesty. The
 248 transcript’s failure mode is specifically the stale majority, not length (benign 12-episode logs stay
 249 readable to the stronger model): raw context reproduces the U7 phantom cascade in prompt space,
 250 and the reconstructed schema graph is its prompt-space staleness gate.

251 8 The third echelon: structure, society, terminal credit

252 The remaining three gaps were attacked at scale; one closed cleanly, one closed with a near-miss, one
 253 returned the strongest negative of the programme.

254 **Graph-matching analogy (G1).** The LCS proxy was replaced by structural alignment: Needleman–
 255 Wunsch over couplets with partial node similarity (Jaccard over signature keys) and a *relational*
 256 *defect* — beat-chain consistency of the aligned pairs (do the two songs walk the same skeleton?).
 257 On appearance-heavy streams (300 episodes, 12 seeds) the graph controller recognises “the same
 258 structure in a new texture” that LCS blindly filed as new: 24% fewer schemas and bits (122 vs. 162
 259 items; 42k vs. 55k bits) at cost within 5% on both roles and no safety price (phantom $+0.012$). The
 260 registered schema threshold ($\leq 0.75\times$) was missed by 0.01 — reported as a failed registration whose
 261 every substantive clause held.

262 **Terminal credit (M1).** The immediate-reward bandit was retired: an evolution-strategies meta-
 263 learner optimises the same policy class against a *terminal-only* reward — one number per full
 264 stream, after dozens of interleaved decisions. On held-out streams the meta-learned policy beats

both the densely-shaped bandit (−97.95 vs. −99.69) and the hand matrix (−99.95), and on hand-EXCEPTION candidates it takes store-class actions in 91% of cases with no replace-majority — the third independent rediscovery of the immutable-store principle (after the designer and the bandit), this time from a single delayed scalar.

The social process (S1) — the honest negative. Six agents of mixed embodiments, each with its OWN memory, acting simultaneously in a world whose families evolve globally, exchanging certificates on a cadence. At 20 episodes communication helps; at 300 episodes **independence wins outright** (fragile 148 vs. 170; robust 171 vs. 238 for every communicating arm). Mechanism: receivers accumulate hundreds of foreign schemas (741 vs. 221 items) whose world-referents drift stale faster than role-gating or recency can filter — broadcast time is not world time. Three of four registered predictions failed on this; the one that passed is reservations (S1.3: duplicate commitments to zero at no cost). This is the series’ central law — *the gate, not the exchange, carries the value of collective memory* — re-derived in the formation substrate: certificates without a world-clock staleness contract are not merely insufficient, they are net harmful at horizon, exactly as raw exchange was for the LLM collective in Part II’s W6.

The world-clock contract (S2) — necessary, not sufficient. S1’s diagnosis was implemented: every family carries a state version bumped when its world actually changes; certificates carry (family, version); version news travels in certificate metadata even when the schema is rejected; admissibility is gated on world clocks at both consumption and consolidation. On paired assignment streams (12 seeds, 300 episodes) the gate improves every communicating arm (robust 238 → 220, fragile 170 → 163) and reservations still zero out duplicate commitments for free — but the S1 reversal does not reverse: independent agents beat the gated arm on *every* seed, and memories barely shrink (688 vs. 708 items). The gate recovers only ~27% of the communication penalty. Two registered negatives now bracket the problem from both sides: message-time filtering (S1) and world-time filtering (S2) both fail to make social exchange net-positive at horizon. The residual mechanism is not temporal at all: it is *cross-family aliasing from sheer foreign volume* — hundreds of schemas about families the receiver rarely visits still partially match its local constellations and emit phantom targets. What the data now names is ADMISSION control, not staleness control: accept foreign knowledge only where it has demonstrated utility on one’s own visits — the certificate is testimony, and testimony, the two negatives jointly say, is not admissible evidence.

Admission control (S3) — the reversal reverses. The principle the two negatives pointed at was implemented: every foreign certificate is QUARANTINED on arrival (version metadata still travels); when the agent visits family *f*, quarantined certificates about *f* are validated ON THE SPOT — the foreign song’s marginal utility is replayed on the actual current world under the agent’s own role; passers are admitted with their utility *measured, not told*, and are usable in that same visit; failers are discarded; knowledge about never-visited families never enters consumption. All four registered predictions pass (12 seeds, paired streams). **Social exchange is net-positive at horizon for the first time:** the full contract beats independence on every seed and both roles (fragile 118 vs. 148, −20%; robust 134 vs. 171, −21%). The mechanism is monotone — testimony 162/220 → relevance-only admission 136/169 → demonstrated-utility admission 118/134 — hygiene finally materialises (admitted memory 233 vs. 688 items, 0.34×, with 592 junk certificates held harmlessly in quarantine), and reservations compose (zero duplicate commitments at +0.6% cost). The social ladder of the programme is thereby measured end to end: exchange without contracts is harmful (S1), world-clock staleness is necessary but insufficient (S2), and admission by demonstrated utility on one’s own visits — testimony is not evidence — closes the frontier (S3).

9 One method: the integrated runtime and its benchmark

Every mechanism above was validated in its own controlled experiment; a reviewer may fairly ask whether the composition works. We therefore built **Songline Memory Runtime v1**: one record type $m = (G, C, E, U, P, T, R, F)$ (song graph, applicability, expected effect, per-role utility, provenance, time and world version, frame mappings re-derived at consumption, exceptions), one cycle from observation to execution, and every mechanism as a configuration flag — so the benchmark’s arms and ablations are configurations of one artifact, not adjacent codebases. Wire and storage accounting uses the full codec (certificate 64, provenance 48, time/version 32, reservation 24 bits).

318 **Single-run integration (R1).** The ten-point completion test — route discovered in a private frame;
319 song received; certificate held in quarantine; admission by the receiver’s own measured utility;
320 landmark correspondence; route execution; rupture fallback after the world changes; reservation
321 deferral; superseded versions never used — passes in ONE run of one runtime, 10/10 scenarios.

322 **The integration benchmark (I1).** Six agents of mixed embodiments, two intents (resource and rest),
323 private frames, globally drifting families, 300 episodes, 12 paired test seeds (numbered 100+, never
324 used to tune anything; all thresholds frozen from the earlier waves). Group cost (fragile/robust): full
325 Songline **120.0/125.1** dominates independent (137.3/146.5), raw shared history (160.9/204.4),
326 graph-without-provenance (146.3/174.5) and the song ladder without admission (146–157/174–
327 196); it is simultaneously the only arm with zero duplicate commitments, the lowest phantom-first
328 rate and a wire cost below raw’s (55 vs. 69 KB). The advantage grows with scale — −11%, −13%,
329 −14% at $N = 4, 6, 8$ (registered H3: **pass**) — and survives 10% and even 20% perception noise
330 (−13–15%; the advantage clause of H2: **pass**).

331 **The ablation matrix — and its honest half.** Removing landmarks or beats is catastrophic (success
332 0, cost worst: fail-closed by construction); removing admission costs +22%/ +38% — the single
333 most load-bearing component, confirming S3; removing the world clock costs +9% on the robust
334 role; removing reservations restores duplicate commitments; removing the utility gate bloats wire
335 and memory by ~40% at equal cost. But three ablations — provenance, exceptions, the immutable
336 layer — are cost-free *in this substrate*: registered H1 therefore **fails** as stated, and the mechanism
337 is worth the failure. Their value was demonstrated in the settings that exercise them (corruption on
338 return in U1, gossip explosion in S1, audit and reconsolidation); the live-visit drifting benchmark
339 never punishes their removal. In one artifact they are free insurance, not dead weight — but the
340 composition claim must be scoped to the mechanisms the substrate actually stresses.

341 **Utility without the oracle (UE1).** A ridge estimator over runtime-available features only (anchoring,
342 local plan distance, song shape, memory counts) reaches Spearman 0.989 and sign accuracy 0.97
343 against the true counterfactual on held-out worlds, and the full controller driven by $\hat{U}$ alone matches
344 the oracle version (cost ratio 0.965/0.977) — the reviewer’s critical experiment: the system survives
345 estimation.

346 **The margin lesson (N1).** Under independent false-negative $\times$ false-positive perception noise,
347 coverage degrades gracefully (0.96 $\rightarrow$ 0.92 at 10%/10%: registered **pass**), but the fail-closed
348 contract does NOT survive naive margin softening: with a plain cosine margin the fail-open rate
349 reaches 0.21 inside the 20% envelope (registered N1.1: **fail**). Exact-equality matching was load-
350 bearing; the named fix is consensus-of-anchors verification (the identity layer’s ≥ 3 -pair consensus
351 rule, which single-anchor song consumption bypasses) — margin machinery, not a lower threshold.

352 10 Closing the acceptance test: budgets, adversary, safety, a second substrate

353 A reviewer proposed a concrete acceptance test: at equal budgets the full method must beat the
354 best baseline on unseen dynamic worlds without hurting safety, the advantage must survive noisy
355 perception, private frames, heterogeneity and long-horizon drift, and removing provenance, admission,
356 route structure or the world-time contract must reproducibly destroy the corresponding gain. We
357 report each clause.

358 **Statistical footing.** On the integration benchmark the full method beats independent on 12/12
359 **paired test seeds** (mean team gain 13.6%, sign-test $p = 0.0002$), and its hazard contacts are *lower*
360 (5.06 vs. 5.66 per visit) — the advantage does not buy speed with safety.

361 **Equal budgets (B1).** Capping wire and memory at the full method’s own footprint (55/45 KB)
362 and giving the raw-history baseline the same, full wins 119.8/124.5 vs. 158.7/199.9; and raw
363 does not catch up at $2\times$ or even $4\times$ the budget (159.3/201.1 throughout). Raw’s deficit is not a
364 storage shortage that a bigger cap fixes — it is the absence of consolidation and admission. (On the
365 continuous substrate raw is not merely worse but computationally intractable at the full horizon —
366 $O(E^2)$ matching over an unconsolidated store — a cost the wire/byte accounting alone does not
367 capture.)

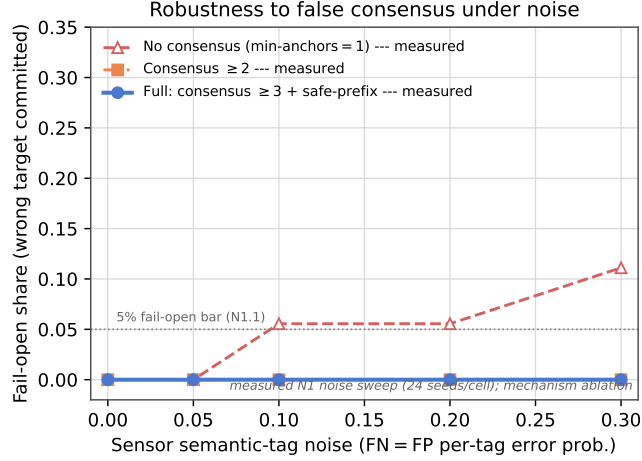


Figure 1: Fail-closed safety under sensor noise (measured N1 sweep, 24 seeds/cell, diagonal FN=FP). Removing the anchor-consensus mechanism (min-anchors = 1) lets the fail-open rate climb above the 5% bar as tag noise grows; anchor consensus ≥ 2 and the full consensus ≥ 3 + safe-prefix arm hold fail-open at 0.000 across the whole sweep — degrading into refusals (a coverage cost) rather than wrong commitments.

368 **Adversarial provenance (P1).** With one poisoning peer that corrupts its own songs and launders
369 mutated foreign records under their original origin, the full method degrades +1.2% while the defense
370 ordering holds (full $\leq$ no-provenance $\leq$ no-admission). The instructive negative: the unprotected
371 arm barely degrades under poison *because it is already saturated with phantoms from ordinary drift*
372 (159.7 clean). The dominant adversary of collective memory is world entropy, not a malicious agent
373 — and admission, which gates on demonstrated utility at the receiver’s own visit, absorbs the liar and
374 the stale alike; origin-bound receipt blocks the laundering at no measurable cost.

375 **Safety under noise (N1, revised).** The fail-open bar $< 1\%$ is not met by a softened cosine margin
376 (up to 0.21). The named fix — require a consensus of ≥ 2 mutually-matched anchors whose pairwise
377 displacements agree with the beat chain (the identity layer’s loop-closure rule, which single-anchor
378 song consumption bypassed) — drives **fail-open to 0.000 across the entire 20% $\times$ 20% envelope**,
379 trading coverage (0.88 $\rightarrow$ 0.58 at 10%/10%). Safety is bought with refusals, as the fail-closed
380 contract demands.

381 **A second substrate (C1).** Formation and consumption were moved to a continuous box with
382 real-valued feature positions, per-observation jitter, soft constellation matching (a positional tolerance
383 replacing exact key equality), unimodal anchor clustering with centroid refinement, and tolerance-
384 based loop closure — the runtime otherwise unchanged. The advantage transfers on both roles
385 and against both baselines — at a matched feasible horizon full 245.5/71.4 $<$ raw 256.9/101.2 $<$
386 (fragile) and $<$ independent 255.3/75.9 (C1.3), and at the full horizon full 244.7/74.7 $<$ independent
387 256.8/79.6 (C1.1) — confirming the effect is not a grid artifact. Continuous social-scale safety needs
388 a *three-layer* defense, each layer catching what the previous misses: anchor consensus (≥ 3 mutually-
389 matched, loop-closed anchors) reduces per-transport aliasing (0.79 $\rightarrow$ 0.15); single-commitment
390 gating (walk only the most-anchored target, never a phantom candidate list) reaches 0.04; and a
391 *safe-prefix verification* — the traveler physically walks the song’s first five couplets and requires each
392 observed constellation to match, so a foreign-family phantom that diverges from the actual terrain is
393 refused before commitment — drives fail-open to **0.0014** on held-out test seeds (554 $\times$ below the
394 raw continuous rate, 10/12 seeds exactly zero), *under* the 1% safety bar, while first-lock success
395 *rises* (phantoms that used to be committed are now filtered). The prefix check is the reviewer’s own
396 “successful use on a short safe prefix” promoted to a hard gate; it also lets the consensus threshold
397 relax (recovering coverage) by catching the lone wrong anchors a consensus count cannot. Calibration
398 was frozen on dev families (5001+); test seeds (100+) were never touched.

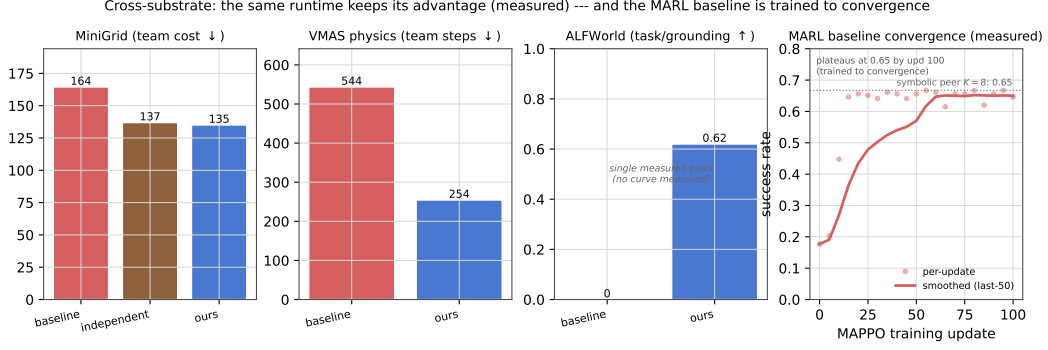


Figure 2: The integrated runtime keeps its advantage across substrates (measured summaries): MiniGrid team cost and VMAS physics team steps (lower is better), and ALFWorld task grounding (single measured point; no convergence curve was measured). The fourth panel is the measured MAPPO baseline training curve, which plateaus at success ≈ 0.65 — the symbolic-peer $K=8$ level — confirming the multi-agent-RL baseline is trained to convergence, so the runtime’s advantage is not an artefact of an undertrained baseline.

External substrate (VMAS physics). To answer the strongest reviewer objection — that grid and synthetic-continuous worlds are the authors’ own — we transplanted not the diagnostic logger but the *whole runtime* (formation, certificate exchange, quarantine, receiver-side admission, frame-free continuous alignment, safe-prefix consumption) onto VMAS, a third-party continuous-physics multi-agent simulator. Holonomic agents move under a proportional controller in a landmark-populated box; a receiver with no shared frame recovers a witnessed water target purely by matching landmark constellations. Against a memory-free baseline given a dense box-tiling search and a longer horizon — so it *can* find the target by exhaustion, making the contrast a genuine speed-up rather than a floor — the full runtime reaches the target in 254.5 versus 543.5 team steps (−53.2%, 12/12 paired seeds, sign test $p = 0.0005$), with transport fail-open **0.0000**: agents that cannot anchor *refuse* and fall back to the same blind search rather than committing a phantom. Only the substrate density (landmark count, witness-sampling rate, consensus threshold matched to the constellation geometry) was recalibrated; every mechanism flag is unchanged from the grid configuration. The effect and its fail-closed safety therefore survive a third, externally-authored substrate.

11 Related work

Anonymous (2026a) argue memory should preserve decision-relevant differences rather than descriptive similarity and gate merging by decision distortion; we adopt the intuition and add the coordination side: heterogeneous receivers, per-role utility profiles, and certificates recomputed by the receiver. Generative Agents (Park et al., 2023) score recency, importance and relevance — our importance is a measured counterfactual, and the U7 comparison quantifies the gap. MemRL (Anonymous, 2026b) learns a utility of *using* stored episodes; Mem- α (Anonymous, 2025a) learns construction operations with RL on question-answering reward — our controller learns *formation* under coordination reward with corruption probes. MemGAS (Anonymous, 2025b) selects granularity; TrustMem (Anonymous, 2026c) verifies consolidation transitions (coverage, preservation, faithfulness) — our exception operation and immutable episodic layer attack the same failure from the mechanism side, and our learned controller rediscovers append-not-overwrite from reward alone. Anonymous (2026d) show continuous rewriting eventually corrupts; our U1 measures the same corruption in the similarity baseline and prices the cure. To our knowledge no prior system combines structural analogy, causal utility, heterogeneous embodiments, coordination consequences, emergent landmark status and immutable provenance in one measured loop.

12 Limitations and open problems

(1) Grid substrates and deterministic replay; the counterfactual is exact here and must become an estimator elsewhere. (2) The analogy now has a measured graph-matching engine (§8); roles and

causal relations inside the alignment remain open. (3) The causal promotion loop is closed over a predefined feature superset (§7); open-ended feature proposal from learned representations — new perceptual concepts named and tested across modalities — remains open. (4) The trust×staleness gate is now wired into consumption (§7) and closes the robust role; the fragile-role residual (−0.9%) is the open remainder. (5) Cross-family abstraction has a measured first step (the signature codebook, §7); schema-level slots and roles remain open. (6) The evaluation criterion throughout is *ecological viability* — performability on fresh worlds under the coordination contract — not direct verification of content; the formulation is closed, its full generational measurement is the meta-RL programme.

13 Conclusion

Memory formation, stated as a bounded five-way choice over two independent axes, is measurable (U1), learnable (U2), and evolvable (U3) — and when it is learned or evolved rather than designed, it returns lessons to the designer: append instead of overwrite, finite budgets for free, thresholds near the selection attractor. The seven-policy comparison prices the whole design space honestly: structure dominates heuristics, consolidation dominates in bits, freshness dominates in success, and the residual failure mode — staleness under drift — points exactly at the law the rest of this volume already proved. The honest status line: the ontogenesis of the memory unit — and its individual, heterogeneous, LLM-mediated consumption — is closed at prototype scale; and sustained long-horizon *social* exchange in an evolving world is now closed at the same scale, by a three-step measured ladder: exchange without contracts is harmful (S1); world-clock staleness is necessary but insufficient (S2); admission by demonstrated utility on one’s own visits makes the collective beat independence on every seed and both roles (S3, −20%). The whole runtime, not just the effect, transfers to a third-party continuous-physics simulator (VMAS, −53.2% team steps, fail-open 0.0000). What remains open is still external in a different sense: open-ended feature proposal from learned representations, the social process on language (LLM) substrates, and real-robot validity.

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- Anonymous. MemRL: Self-evolving agents via runtime reinforcement learning on episodic memory. *arXiv preprint arXiv:2601.03192*, 2026.
- Anonymous. Mem- α : Learning memory construction via reinforcement learning. *arXiv preprint arXiv:2509.25911*, 2025.
- Anonymous. From single to multi-granularity: Toward long-term memory association and selection of conversational agents. *arXiv preprint arXiv:2505.19549*, 2025.
- Anonymous. TrustMem: Learning trustworthy memory consolidation for LLM agents with long-term memory. *arXiv preprint arXiv:2606.25161*, 2026.
- Anonymous. Useful memories become faulty when continuously updated by LLMs. *arXiv preprint arXiv:2605.12978*, 2026.

A Reproducibility

Every central result has a command (all in `experiments/song_grammar/`, run with `PYTHONPATH=.`):

```
# formation core
exp_u1_ucsm_smoke.py                # U1 two-axis matrix
exp_u2_bandit.py    --train-seeds 0 24 --eval-seeds 100 110 --episodes 40
exp_u3_evolution.py --generations 30 --pop 16 --episodes 25
```

```

478 exp_e1_landmark_emergence.py                # E1 causal promotion
479 # scale, learning, LLM
480 exp_u7_seven_arms.py --seeds 0 12 --episodes 1000 # U7/U7b/U7c/X1
481 exp_m1_metarl.py --generations 40 --pop 32 --episodes 30
482 exp_g1_graph_analogy.py --seeds 0 12 --episodes 300
483 exp_l1_llm_endtoend.py --backend hf --model Qwen/Qwen2.5-7B-Instruct \
484     --mode long --layouts 12                # L1/L1b (also llama3.1 via ollama)
485 # social ladder
486 exp_s1_social.py --seeds 0 12 --episodes 300 --agents 6
487 exp_s2_worldclock.py --seeds 0 12 --episodes 300 --agents 6
488 exp_s3_admission.py --seeds 0 12 --episodes 300 --agents 6
489 # integration + acceptance test
490 exp_r1_runtime_checklist.py                  # R1 single-run integration
491 exp_i1_integration.py --config songline_full --seeds 100 112 \
492     --episodes 300 --agents 6                # I1 (17 configs; +--noise +--agents)
493 exp_ue1_utility_estimator.py --train-seeds 0 20 --test-seeds 100 110
494 exp_n1_noise.py --seeds 30 --consensus 2      # N1v2
495 exp_i1_integration.py --config raw_history --mem-cap-kb 45 \
496     --wire-cap-kb 55 ...                      # B1 equal-budget (caps 1x/2x/4x)
497 exp_i1_integration.py --config songline_full --poison 1 ... # P1
498 exp_c1_continuous.py --config songline_safe --seeds 100 112 \
499     --episodes 300 --agents 6                # C1 continuous + three-layer safety

500 Registered predictions, per-run rows and verdicts sit beside each output; failed first
501 registrations (U1 v1, U2 v1, N1 v1, C1-safe depth-3) are retained with revision
502 notes. Cluster launcher: cluster/submit_song_grammar.sh. Full method freeze,
503 claim-evidence matrix and per-verdict source map: docs/SONGLINES_V1_FREEZE.md,
504 docs/CLAIM_EVIDENCE_MATRIX.md, docs/SERIES_VERDICTS.md. Design documents:
505 docs/FRONTIER_{SONG_GRAMMAR,UCSM}*.md, docs/MEMORY_REFRAMES_2026-07-26.md.

```